Project Report - Day 6

Title

TASK-06: Implementing a Stateful Chat Interface with Context Windows.

Objective

The objective of this task is to improve the chatbot's conversation flow so that it does not begin with direct technical questions. Instead, the bot should first greet the user, ask simple and easy questions one at a time, and if the user responds with "no" or a negative reply, the bot should politely move on to the next question..

Libraries Used

1. **os**
    - Used to handle file paths and access the .env file location.
2. **dotenv**
    - Used to load environment variables from the `.env` file.
3. **Google.genai**
    - Used to load environment variables from the .env file.

Errors

1. **Direct Questioning at Start**

   Issue: The chatbot was starting the conversation by asking questions immediately without greeting the user first.

   Solution: An opening greeting message was added so the chatbot now starts in a more natural and friendly way.

2. **Strict Recruiter Behavior**

   Issue: The system prompt defined the bot as a strict technical recruiter, which made the interaction feel harsh and too formal from the beginning.

   Solution: The system prompt was rewritten to define the bot as a friendly recruiter chatbot with a supportive tone.

3. **Hard Questions Too Early**

   Issue: The bot could begin with technical or difficult interview-style questions too early in the conversation.

   Solution: New prompt instructions were added so the bot starts only with simple questions such as name, interests, skills, or experience.

4. **No Proper Handling of Negative Replies**

   Issue: If the user said “no” or gave a negative reply, the chatbot did not have clear behavior for what to do next.

   Solution: The prompt was updated so the bot politely skips that topic and moves to the next easy question.

5. **API Key Loading Issue**

   Issue: The script may fail if the API key is missing or stored under a different variable name.

   Solution: The code was updated to check the environment variable properly before creating the client.

Output

```
PS D:\Daily_Tasks> & C:\Users\Acer\AppData\Local\Programs\Python\Python310\python.exe d:/Daily_Tasks/Day6_Task/chatbot_day6.py
Recruiter: Hello! Welcome to the recruiter bot. Let's start with a few easy questions. What should I call you?
You: hi my name is samruddhi
Recruiter: It's a pleasure to meet you, Samruddhi! Thanks for joining me today. To get us started, what kind of roles or industries are you currently most interested in?

You: ai intern
Recruiter: That sounds like an exciting field to get into! AI is definitely a hot topic right now. To get a better sense of your background, what programming languages or
tools are you most comfortable working with?

You: python and tools are machine learning
Recruiter: That's a great foundation! Python is definitely the go-to language for AI. Since you've been working with machine learning, have you had a chance to build any p
rojects or complete any coursework using libraries like Scikit-learn, TensorFlow, or PyTorch yet?

You: no
Recruiter: No problem at all! Everyone starts somewhere. Are you currently working on any personal projects or learning anything new in the AI space right now?

You: exit
Goodbye!
PS D:\Daily_Tasks>
```